

APPENDIX

I. THEORETICAL ANALYSIS OF DRGC

Theorem 1. Consider a set of 3D points $\mathbf{P}_i = [X_i, Y_i, Z_i]^\top$ in Euclidean space. These points are projected onto the image plane by a camera with intrinsic matrix $\mathbf{K}$, yielding pixel coordinates

$$\mathbf{p}_i = \pi(\mathbf{P}_i) = \begin{bmatrix} u_i \\ v_i \end{bmatrix} = \begin{bmatrix} f_x \frac{X_i}{Z_i} + c_x \\ f_y \frac{Y_i}{Z_i} + c_y \end{bmatrix}, \quad (1)$$

where f_x, f_y are the focal lengths and c_x, c_y are the principal point coordinates. Applying a small rigid perturbation (rotation and translation) to all 3D points results in

$$\mathbf{P}'_i = \mathbf{R}\mathbf{P}_i + \mathbf{t}. \quad (2)$$

The per-point reprojection error between the perturbed and original projections is

$$\mathbf{e}_i = \mathbf{p}'_i - \mathbf{p}_i, \quad (3)$$

Under small perturbations, the directions of $\mathbf{e}_i$ exhibit global consistency.

Proof: Let the infinitesimal rotation be parameterized by $\boldsymbol{\theta} = [\theta_x, \theta_y, \theta_z]^\top$. Under the small-angle assumption, the rotation matrix can be approximated as

$$\mathbf{R} \approx \mathbf{I} + [\boldsymbol{\theta}]_\times, \quad (4)$$

where $[\boldsymbol{\theta}]_\times$ is the skew-symmetric matrix of $\boldsymbol{\theta}$:

$$[\boldsymbol{\theta}]_\times = \begin{bmatrix} 0 & -\theta_z & \theta_y \\ \theta_z & 0 & -\theta_x \\ -\theta_y & \theta_x & 0 \end{bmatrix}. \quad (5)$$

The perturbed point is then

$$\mathbf{P}'_i \approx \mathbf{P}_i + [\boldsymbol{\theta}]_\times \mathbf{P}_i + \mathbf{t}. \quad (6)$$

Let $\Delta\mathbf{P}_i = [\boldsymbol{\theta}]_\times \mathbf{P}_i + \mathbf{t}$. Now express $\Delta\mathbf{P}_i$ explicitly:

$$\Delta\mathbf{P}_i = [\boldsymbol{\theta}]_\times \mathbf{P}_i + \mathbf{t} = \begin{bmatrix} -\theta_z Y_i + \theta_y Z_i + t_x \\ \theta_z X_i - \theta_x Z_i + t_y \\ -\theta_y X_i + \theta_x Y_i + t_z \end{bmatrix}. \quad (7)$$

A first-order Taylor expansion of the projection gives:

$$\mathbf{e}_i = \pi(\mathbf{P}_i + \Delta\mathbf{P}_i) - \pi(\mathbf{P}_i) \approx \mathbf{J}_{\mathbf{P}_i} \cdot \Delta\mathbf{P}_i, \quad (8)$$

where $\mathbf{J}_{\mathbf{P}_i}$ is the Jacobian of the projection at $\mathbf{P}_i$,

$$\mathbf{J}_{\mathbf{P}_i} = \begin{bmatrix} \frac{f_x}{Z_i} & 0 & -\frac{f_x X_i}{Z_i^2} \\ 0 & \frac{f_y}{Z_i} & -\frac{f_y Y_i}{Z_i^2} \\ 0 & 0 & 0 \end{bmatrix}. \quad (9)$$

Now express $\Delta\mathbf{P}_i$ explicitly:

$$\Delta\mathbf{P}_i = [\boldsymbol{\theta}]_\times \mathbf{P}_i + \mathbf{t} = \begin{bmatrix} -\theta_z Y_i + \theta_y Z_i + t_x \\ \theta_z X_i - \theta_x Z_i + t_y \\ -\theta_y X_i + \theta_x Y_i + t_z \end{bmatrix}. \quad (10)$$

Substituting into the error expression yields the linearized reprojection error:

$$\mathbf{e}_i \approx \mathbf{J}_{\mathbf{P}_i} \cdot \Delta\mathbf{P}_i = \mathbf{J}_{\mathbf{e}_i} \cdot \delta\boldsymbol{\xi}, \quad (11)$$

where $\delta\boldsymbol{\xi} = [\theta_x, \theta_y, \theta_z, t_x, t_y, t_z]^\top$ is the 6D perturbation, and $\mathbf{J}_{\mathbf{e}_i}$ is a 2×6 Jacobian matrix derived from the expansion.

Define normalized coordinates $x_i = X_i/Z_i$, $y_i = Y_i/Z_i$, and assume $f_x = f_y = f$ for easier comparison. The influence of each DoF on the reprojection error is as shown in Table 1. The focal length f is typically large, and the normalized coordinates x_i, y_i are typically within $[-1, 1]$. Therefore, the relative weight from strong to weak is: $w_i(\theta_x) \approx w_i(\theta_y) \gg w_i(t_x) \approx w_i(t_y) > w_i(\theta_z) > w_i(t_z)$.

In conclusion, the reprojection error direction is primarily determined by rotations θ_x and θ_y , which have high weights and consistent influence directions. Translations t_x and t_y further enhance consistency. Although t_z and θ_z introduce less consistency, their weights are significantly smaller. When depth variation is minimal and points are closer to the image center, these inconsistencies can be negligible.

Based on the analysis, when cube projections can be customized, residual direction consistency can be ensured. Specifically, it is not necessary for the directions to be completely identical and maintaining them within a specified angular threshold is sufficient for practical applications.

TABLE I
THE EFFECT OF SINGLE DEGREE-OF-FREEDOM PERTURBATION ON THE RESIDUAL

1 DoF	$\mathbf{e}_i$	w_i
θ_x	$f \begin{bmatrix} -x_i y_i \\ -(1+y_i^2) \end{bmatrix} \theta_x$	$f \sqrt{(x_i y_i)^2 + (1+y_i^2)^2} \geq f$
θ_y	$f \begin{bmatrix} 1+x_i^2 \\ x_i y_i \end{bmatrix} \theta_y$	$f \sqrt{(x_i y_i)^2 + (1+x_i^2)^2} \geq f$
θ_z	$f \begin{bmatrix} -y_i \\ x_i \end{bmatrix} \theta_z$	$f \sqrt{x_i^2 + y_i^2}$
t_x	$\begin{bmatrix} f/Z_i \\ 0 \end{bmatrix} t_x$	f/Z_i
t_y	$\begin{bmatrix} 0 \\ -f/Z_i \end{bmatrix} t_y$	f/Z_i
t_z	$\begin{bmatrix} -f x_i/Z_i \\ -f y_i/Z_i \end{bmatrix} t_z$	$f \sqrt{x_i^2 + y_i^2}/Z_i$

II. SIMULATION EVALUATION

In this section, we designed simulation experiments to quantitatively validate the DRGC theory and to determine the optimal threshold for directional consistency. Specifically, 100 3D points were randomly generated within the space in front of the camera. The camera intrinsic parameters were set with a focal length of 480 pixels and an image plane size of 960×960 pixels. Perturbations were introduced as follows: the rotation error ranged from 0.1° to 4.6° , while the translation error ranged from 0.1 m to 0.5 m. To further analyze the impact

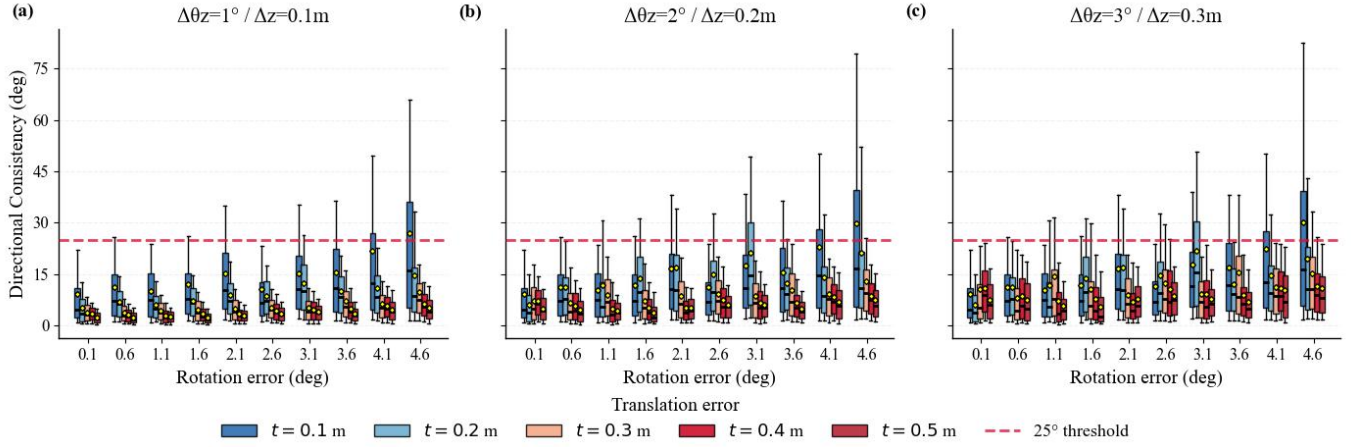


Fig. 1. The simulation results under different levels of perturbation are presented. In particular, we conducted three levels of consistency experiments targeting Z-axis rotation and translation disturbances. The overall results show that, under perturbations of $3^\circ/0.5$ m, the residual direction consistency can be maintained within 25° . This result can serve as a practical threshold for subsequent applications.

of combined errors, we defined three levels of simultaneous rotation and translation perturbations along the Z-axis: $1^\circ/0.1$ m, $2^\circ/0.2$ m, and $3^\circ/0.3$ m. Each set of parameters was run 10 times.

The experimental results of residual direction consistency are presented in Fig.1. Overall, the data reveal an inverse relationship between perturbation magnitude and consistency performance, but the overall angular consistency remains within an acceptable range. Both rotational and translational errors along the Z-axis contribute to increased directional inconsistency. Nevertheless, the simulation results demonstrate robust performance even under substantial perturbations, such as a 3° rotation and 0.5 m translation, with the consistency angle remaining below 25° . Notably, this tolerance is sufficient to accommodate the precision typically estimated during coarse optimization stages. These findings confirm the practical applicability of the proposed DRGC theory in realistic scenarios.

III. COMPLEMENTARY ABLATION RESULT

As shown in the added results in Table II and Fig. 4, our proposed DRGC method achieves significantly tighter inlier selection and lower errors compared to standard RANSAC. Specifically, DRGC drastically reduces the RMSE from 27.7 cm / 2.82° (RANSAC) to 7.8 cm / 1.05° , representing a reduction of over 71% in translation error and 62% in rotation error. Similar substantial improvements are observed across Mean and Median metrics. Furthermore, the maximum error (Max) drops from 73.0 cm to 42.9 cm, demonstrating DRGC's superior robustness against extreme outliers. These results demonstrate our work is a strong methodological framework.

TABLE II
COMPARISON OF RELOCALIZATION ERROR
(TRANSLATION(CM)/ROTATION($^\circ$))

Method	RMSE	Mean	Median	Min	Max
Coarse	38.5/5.45	22.4/2.97	18.2/2.15	10.5/1.07	76.1/8.08
RANSAC	27.7/2.82	17.6/2.18	12.6/0.91	4.8/0.35	73.0/5.97
DRGC	7.8/1.05	6.5/0.76	4.5/0.42	1.9/0.13	42.9/4.24

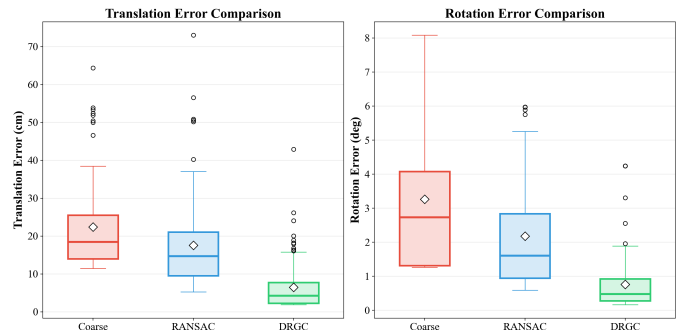


Fig. 2. The ablation results on the different method.